

Correlation of Mathematical Model with AFRL RC-19 Aerothermoelastic Experiment Including an Impinging Shock

AIAA AVIATION Forum 2025

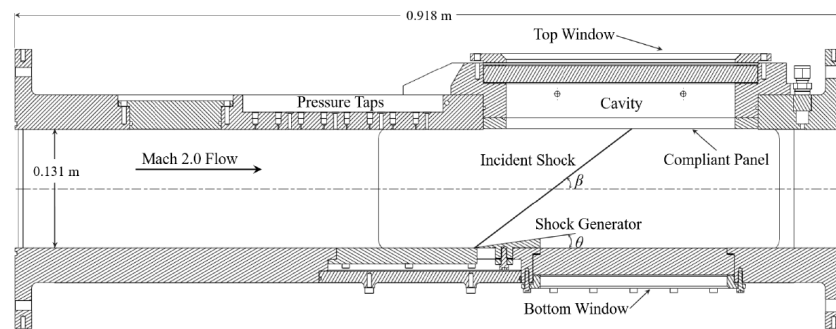
Luisa Piccolo Serafim and Earl Dowell – Duke University

Kirk Brouwer – AFRL, Wright-Patterson AFB

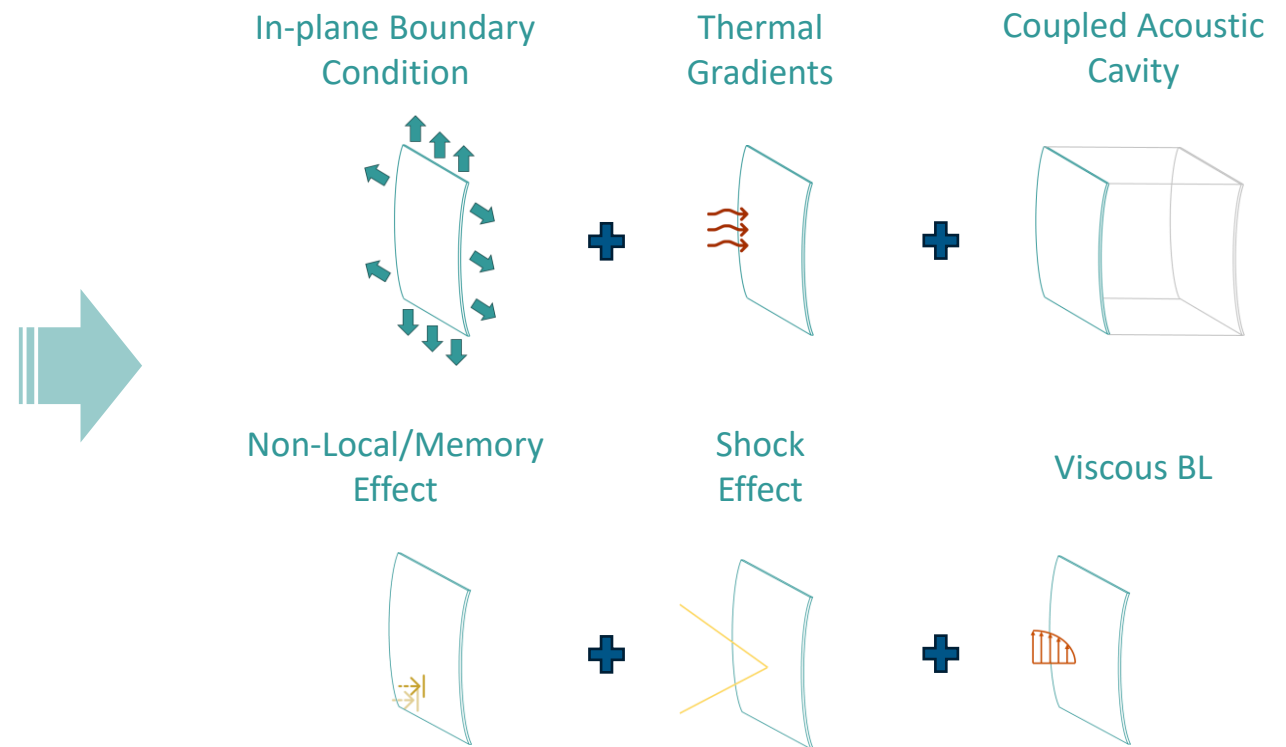
July 22, 2025 | Las Vegas, NV

Motivation for this Work

- To provide the highest possible fidelity in the computational model at an affordable cost; orders of magnitude reduction in cost compared to traditional CFD/CSD methods
- To explore a wide range of relevant parameters including M_∞ , Re , static pressure differential, thermal stresses and structural boundary conditions, both out of plane and in plane.
- **To correlate computational results with experimental results and assess the sensitivity of these results to uncertainties in key parameters**

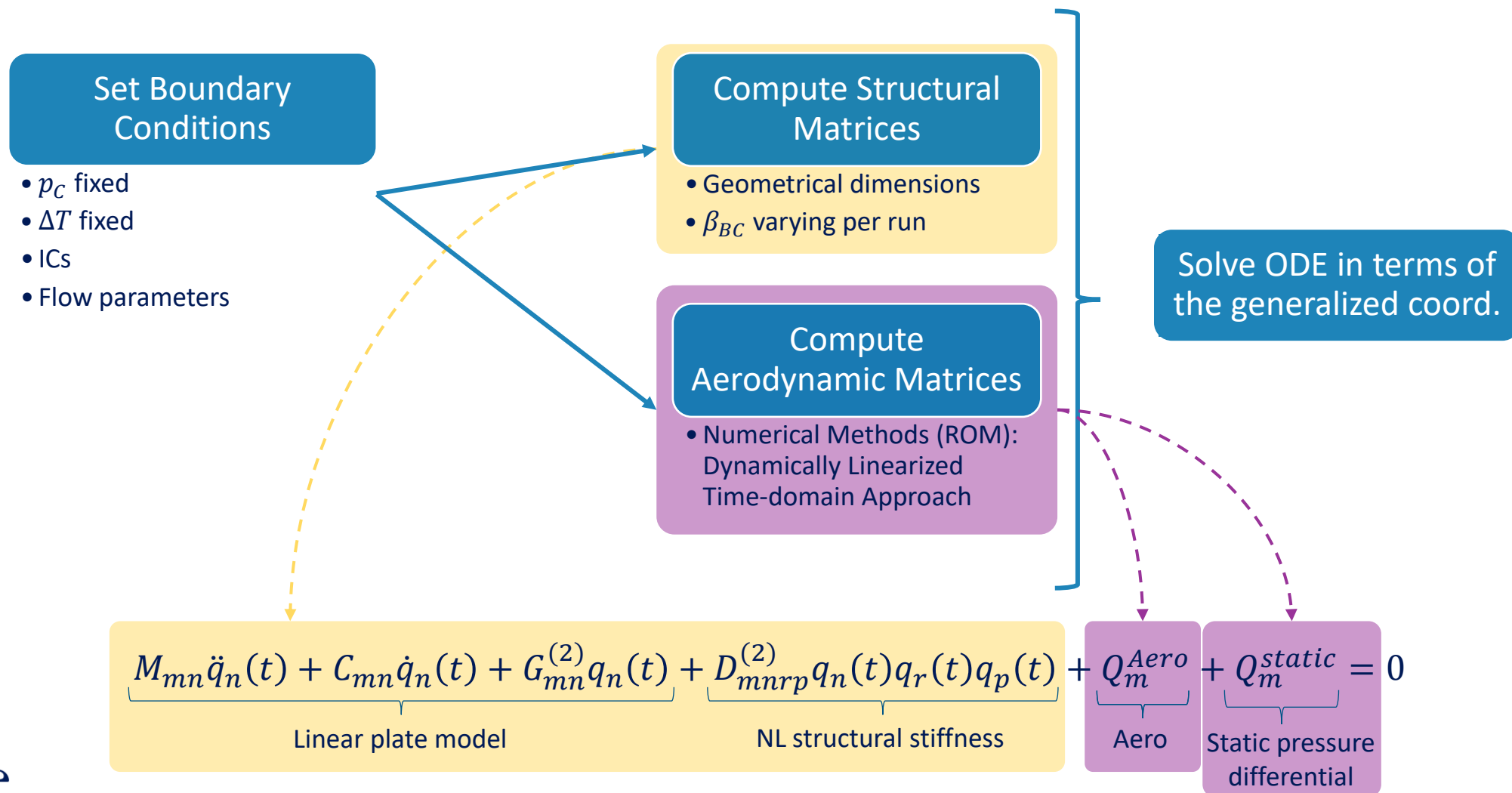


Brouwer et al. AIAA (2021)



Mathematical/Computational Modeling

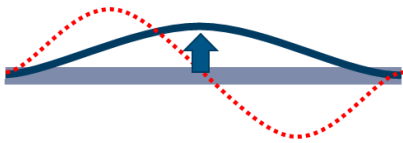
Nonlinear Aeroelastic Solver



ROM to include unsteady aerodynamics into the aeroelastic solution

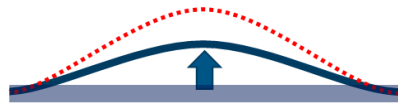
Step change in the CFD domain for each ψ_{mn} ...

... in the panel deformation:



1st set of CFD runs

... in the velocity of the panel deformation:



2nd set of CFD runs

... to obtain a pressure field (output) from a known deformation (input)

$$Q_{mn}^{CFD}(t) = \iint \frac{\Delta p_m(x, y, t)}{\rho_\infty U_\infty^2} \psi_n(x, y) dx dy$$

Work the other way around to obtain the A_{mn} , B_{mn} , and $E_{mn}(t)$ using the input/output relationship

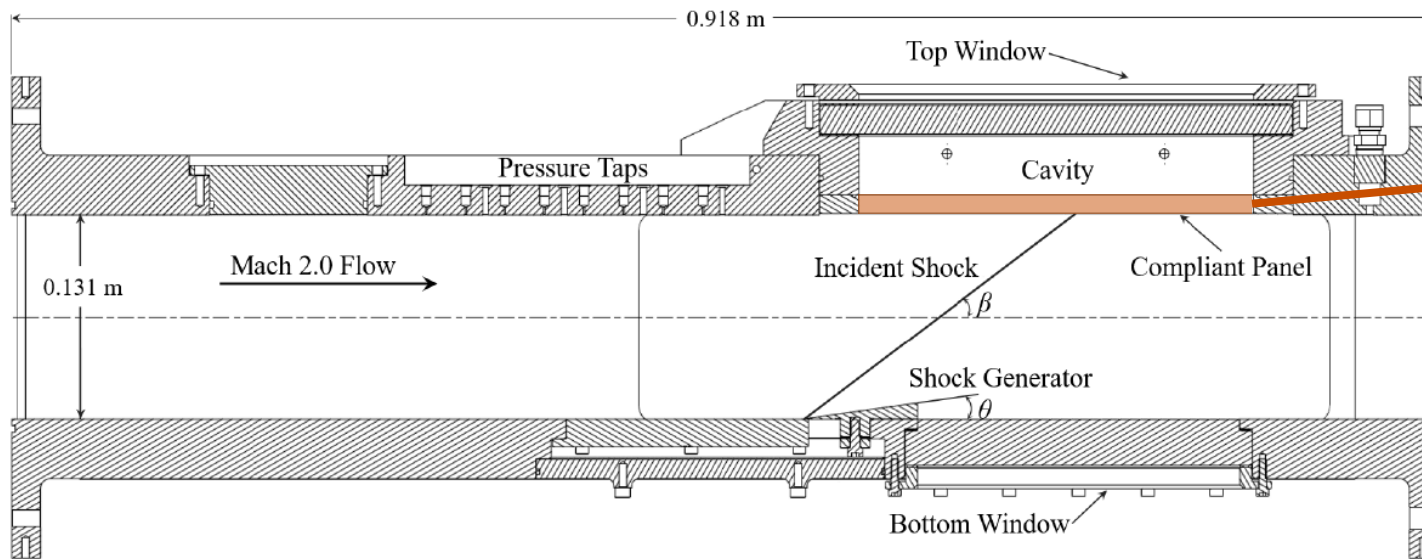
$$Q_{mn}^{CFD}(t) = q_m(t)A_{mn} + \dot{q}_m(t)B_{mn} + \int_0^t q_m(\tau)E_{mn}(t - \tau)d\tau$$

Once A_{mn} , B_{mn} , and $E_{mn}(t)$ are obtained, we can reconstruct the Generalized Aerodynamic Force inside the Aeroelastic Solver for any arbitrary panel deformation ($q_m(t)$, $\dot{q}_m(t)$)

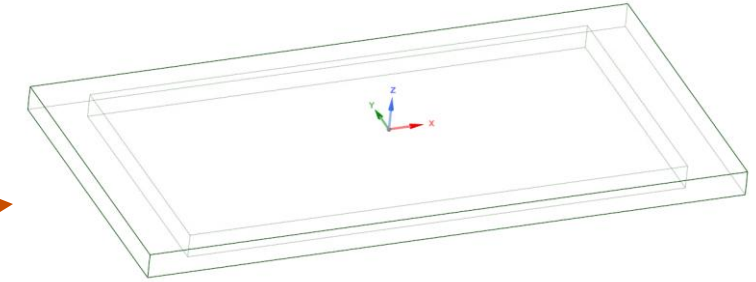
$$Q_{mn}(t) = q_m(t)A_{mn} + \dot{q}_m(t)B_{mn} + \int_0^t q_m(\tau)E_{mn}(t - \tau)d\tau$$

Experimental Study Case: AFRL/SD RC-19 Wind Tunnel Section

Additional Considerations: In-plane boundary condition



Brouwer et al. AIAA J (2021)



Nonlinear structural behavior

In-plane boundary stiffness
 “allowing the panel to deform at the edges as a reaction to the fixed support”

In-plane boundary stiffness

Nondimensional in-plane boundary stiffness

$$\beta_{BC} \equiv \frac{K_{BC} a}{Eh}$$

Panel length a
Panel thickness h



$\beta_{BC} \approx 0$: Free edge displacement
 $\beta_{BC} \gg 0$: No edge displacement

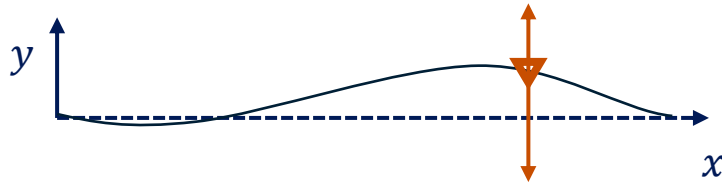
Experimental Study Case: AFRL/SD RC-19 Wind Tunnel Section

Additional Considerations: In-plane boundary condition

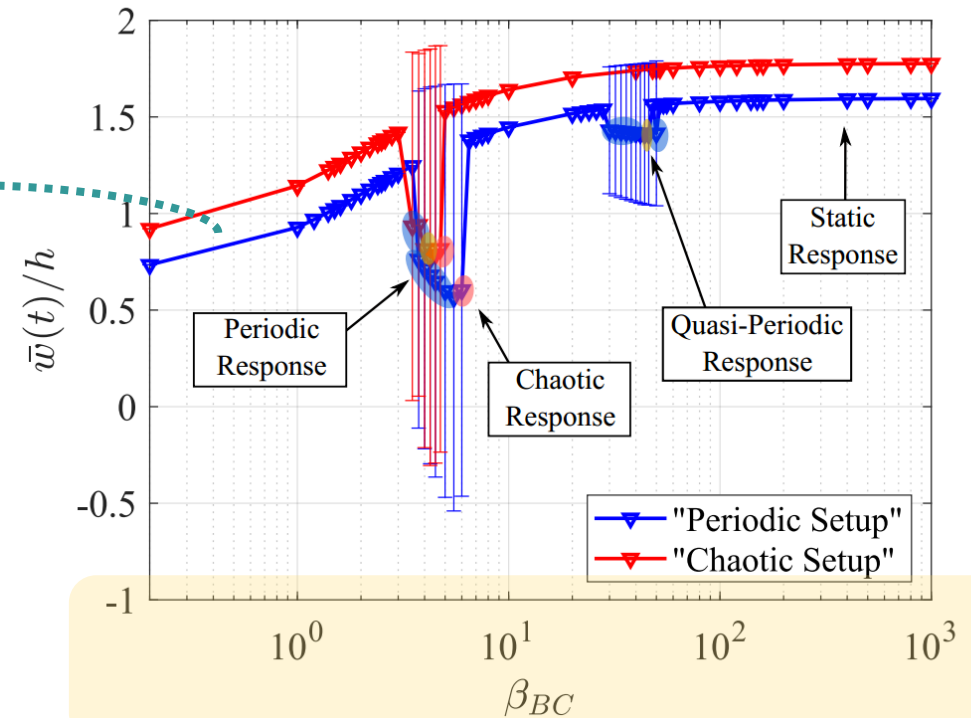
Results are presented as a function of the β_{BC} , which is a **structural** parameter

For example:

Mean and STD of the panel deformation at $\frac{3}{4}$ of the panel length using Linear Piston Theory as the aerodynamic model



Mean & STD deformation

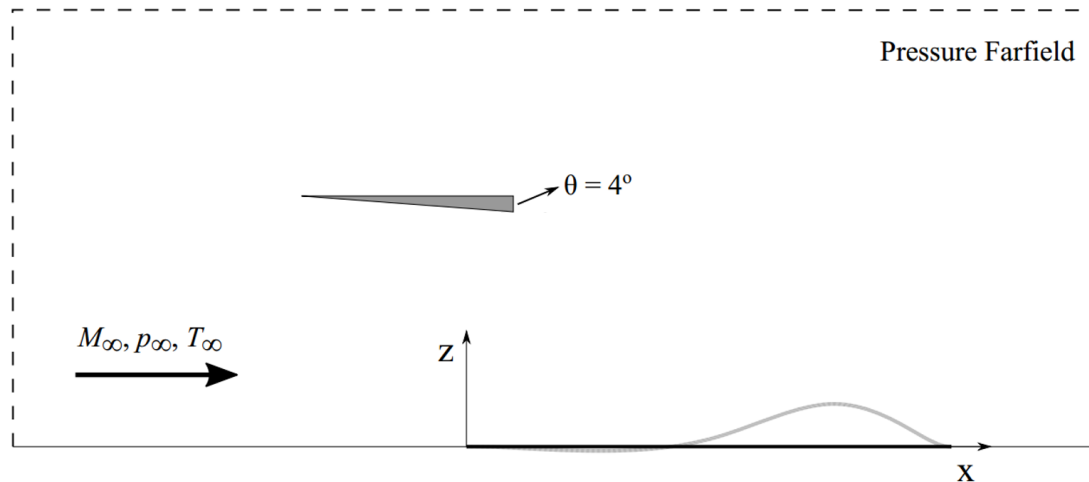


DLTA with a Shock Impingement

$\theta = 4^\circ$ wedge shock configuration

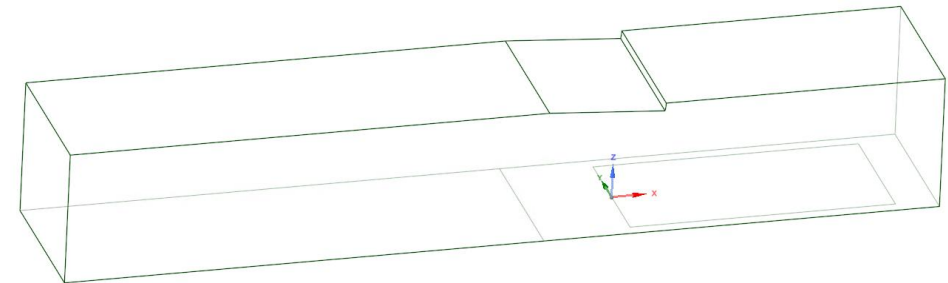
Aerodynamic Model: Euler (unsteady)/DLTA

(Dynamically Linearized Time-Domain Approach)



Aerodynamic Model: RANS (unsteady)/DLTA

(Dynamically Linearized Time-Domain Approach)



In-Plane boundary stiffness sensitivity

$$M_\infty = 2.0$$

ΔT between panel and frame

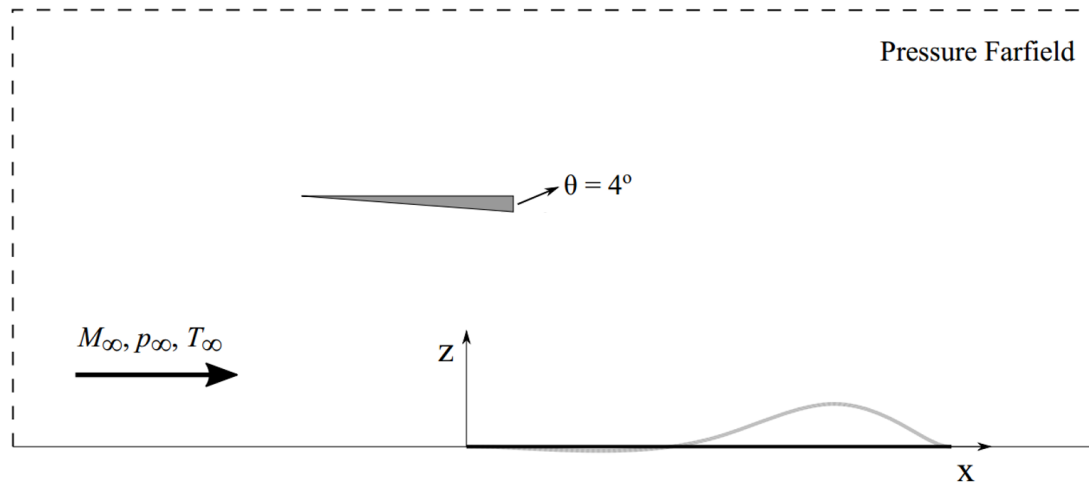
Δp between fluid and acoustic cavity

DLTA with a Shock Impingement

$\theta = 4^\circ$ wedge shock configuration

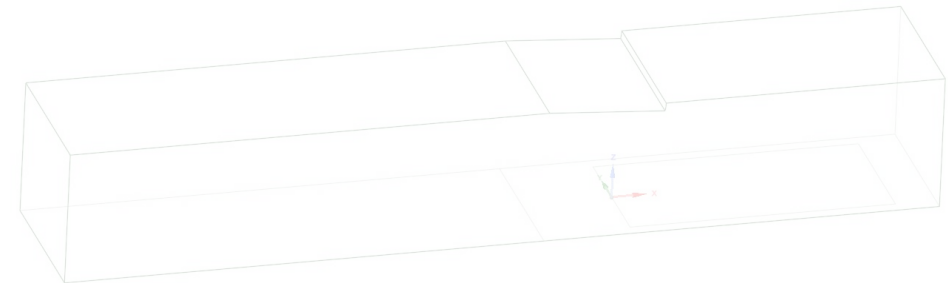
Aerodynamic Model: Euler (unsteady)/DLTA

(Dynamically Linearized Time-Domain Approach)



Aerodynamic Model: RANS (unsteady)/DLTA

(Dynamically Linearized Time-Domain Approach)



In-Plane boundary stiffness sensitivity

$$M_\infty = 2.0$$

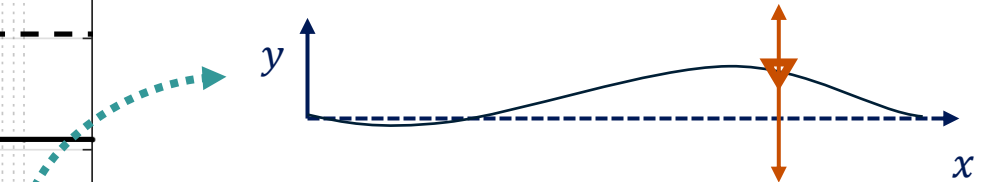
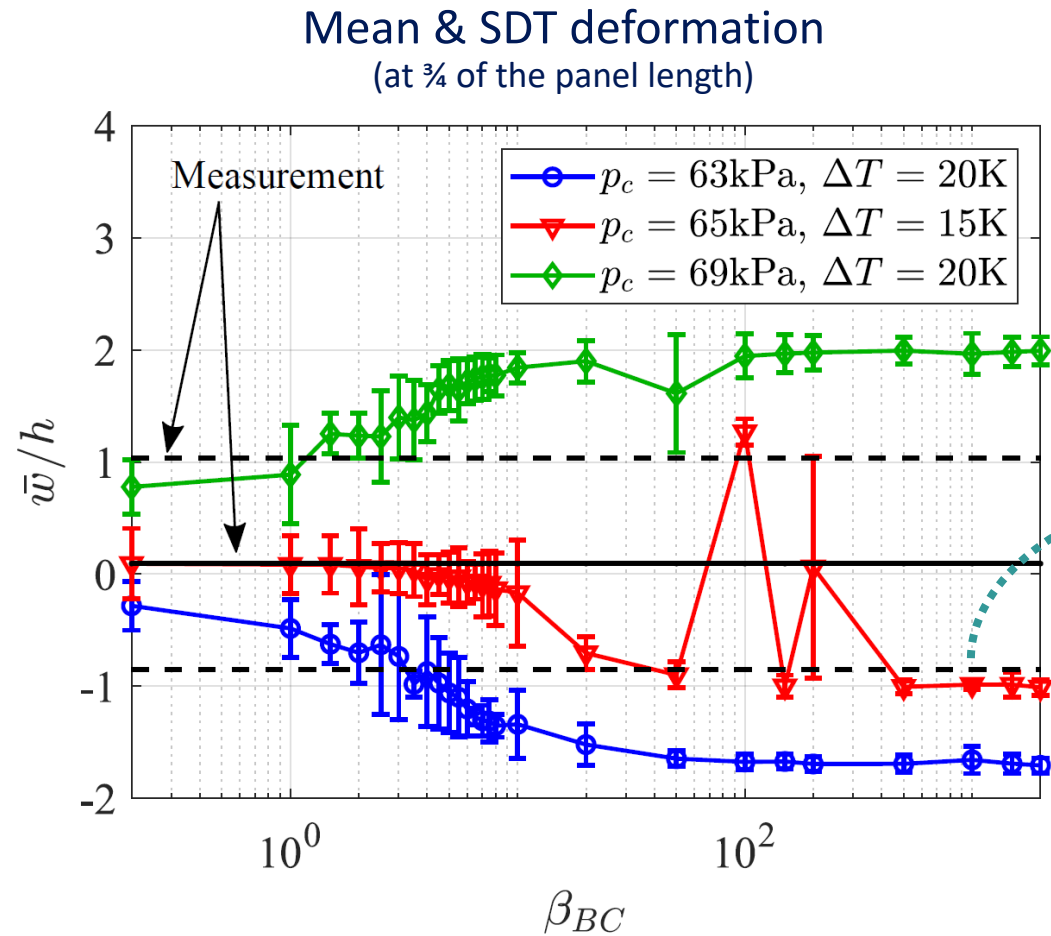
ΔT between panel and frame

Δp between fluid and acoustic cavity

p_c (kPa)	ΔT (K)
68.9	13.3

DLTA with an Inviscid Shock Impingement

$\theta = 4^\circ$ wedge shock configuration

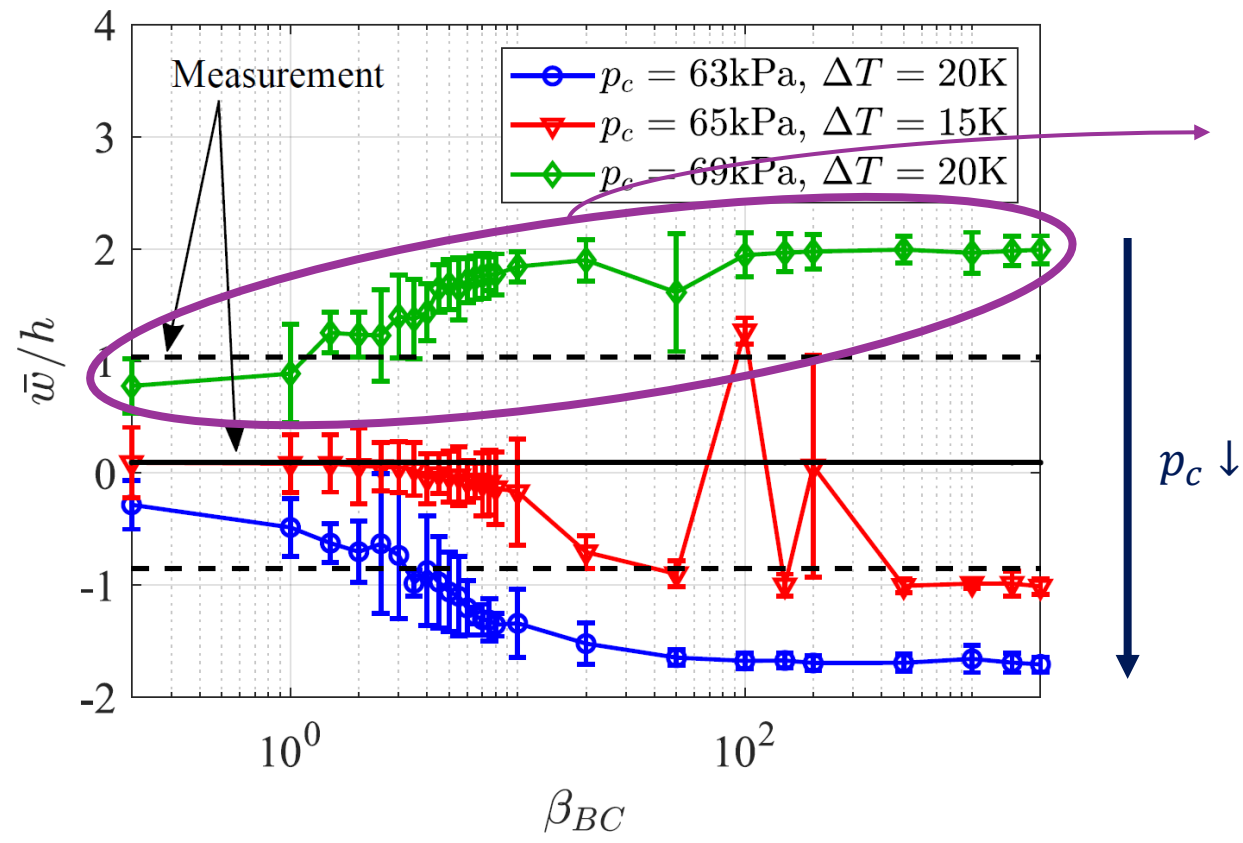


p_c (kPa)	ΔT (K)
68.9	13.3

DLTA with an Inviscid Shock Impingement

$\theta = 4^\circ$ wedge shock configuration

Mean & SDT deformation
(at $\frac{3}{4}$ of the panel length)

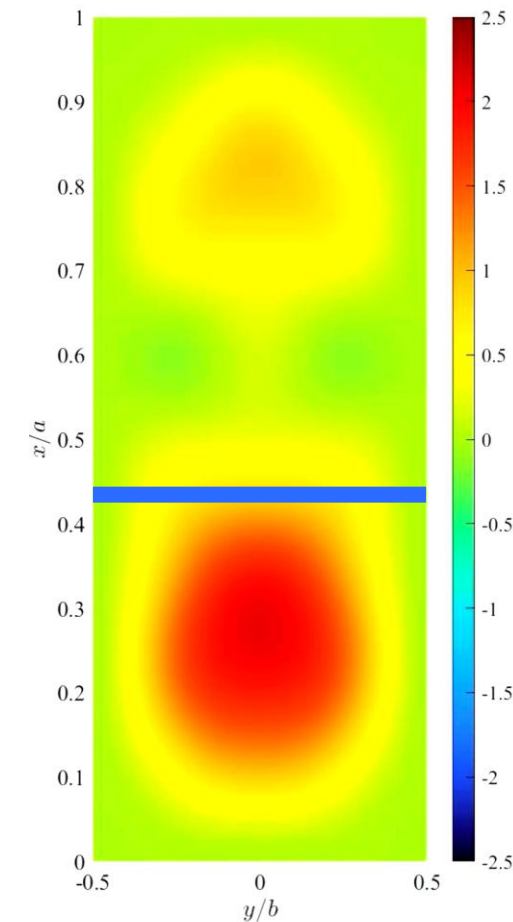
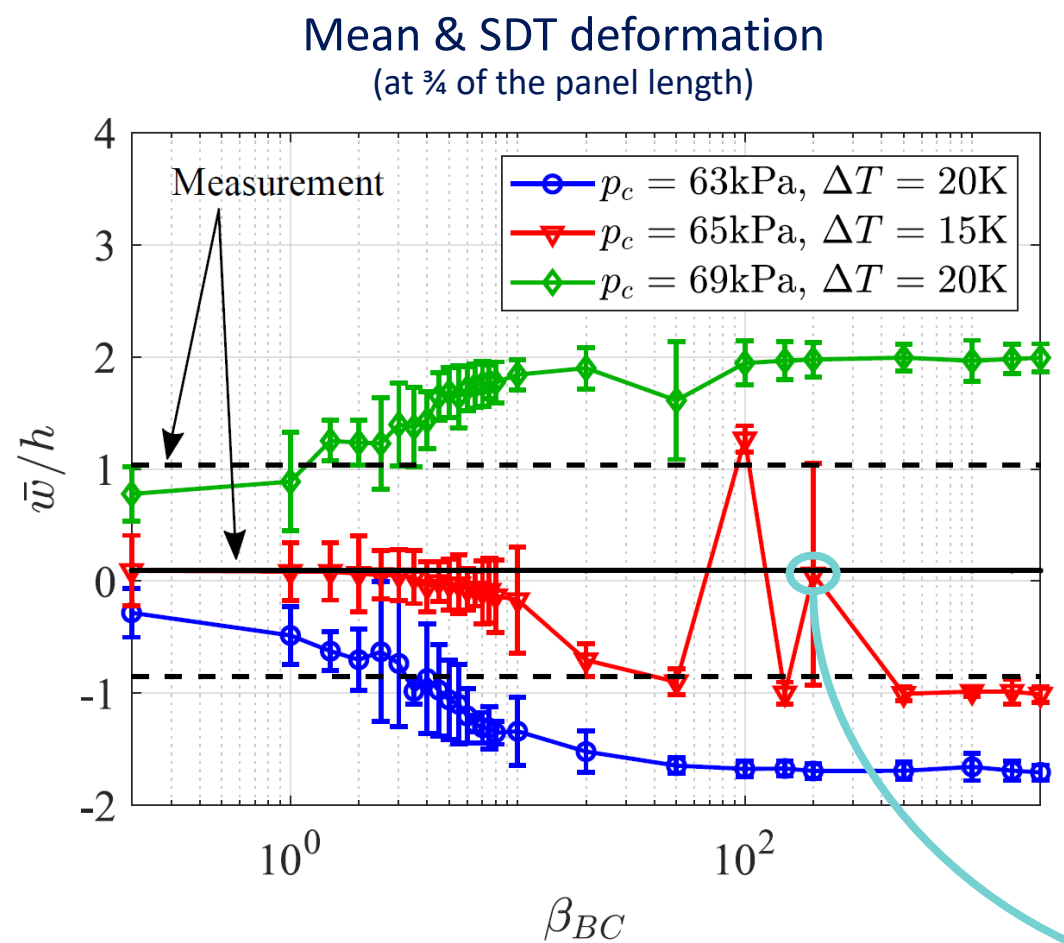


Similar to the wind tunnel setting

p_c (kPa)	ΔT (K)
68.9	13.3

DLTA with an Inviscid Shock Impingement

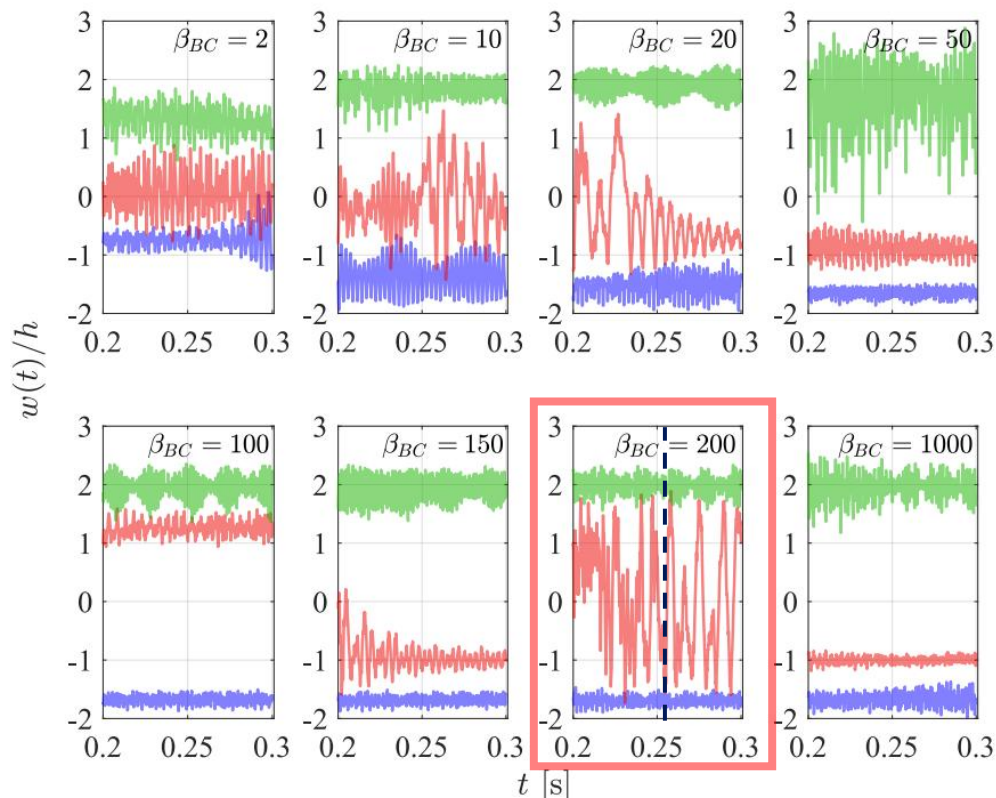
$\theta = 4^\circ$ wedge shock configuration



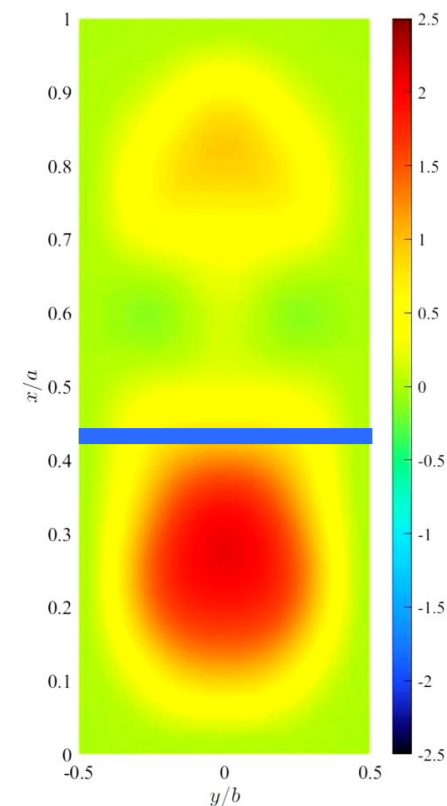
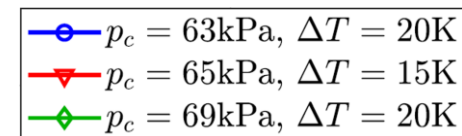
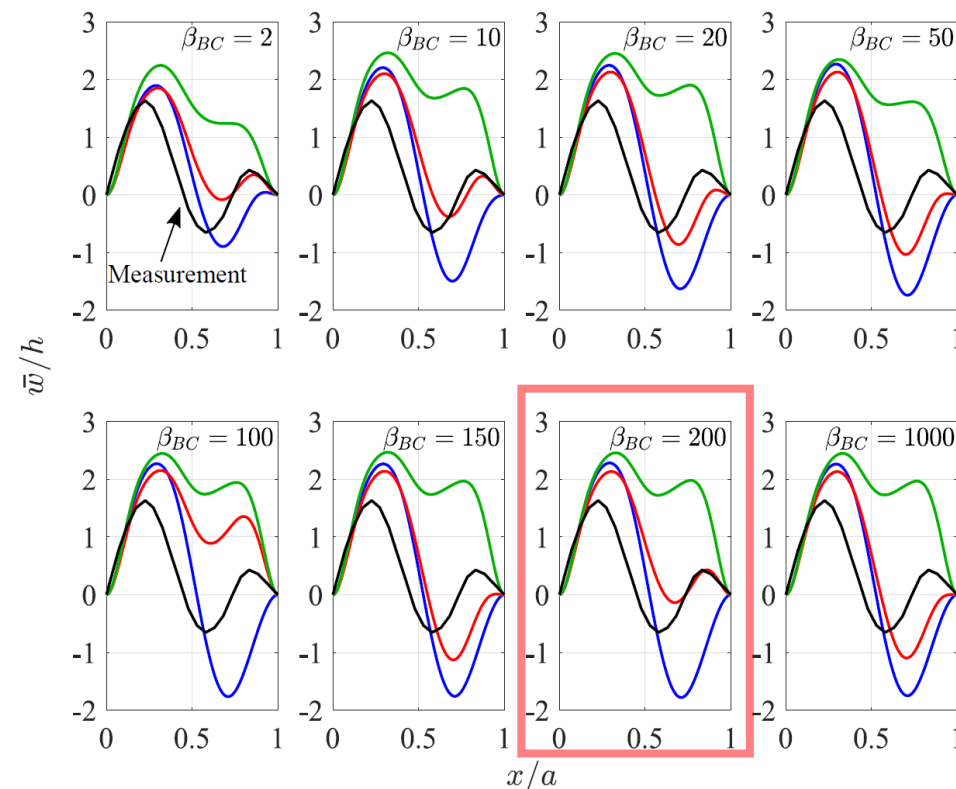
Flow Field

DLTA with an Inviscid Shock Impingement

Deformation in time
(at $\frac{3}{4}$ of the panel length)



Mean deformation



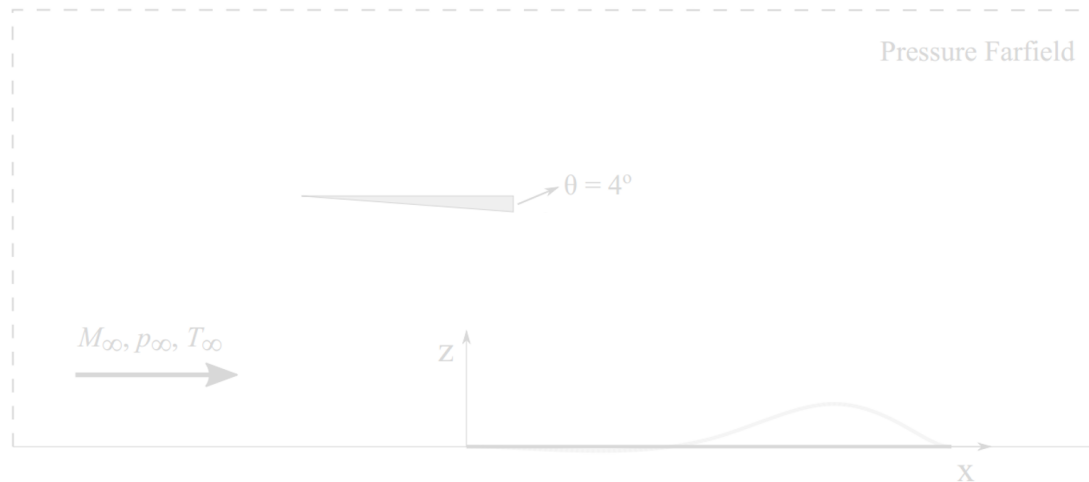
Flow Field

DLTA with a Shock Impingement

$\theta = 4^\circ$ wedge shock configuration

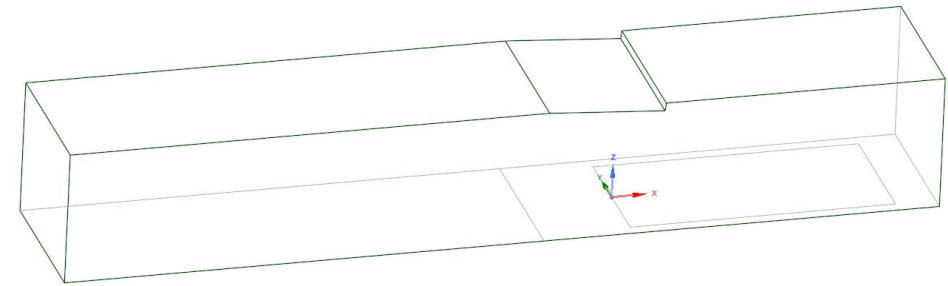
Aerodynamic Model: Euler (unsteady)/DLTA

(Dynamically Linearized Time-Domain Approach)



Aerodynamic Model: RANS (unsteady)/DLTA

(Dynamically Linearized Time-Domain Approach)



In-Plane boundary stiffness sensitivity

$$M_\infty = 2.0$$

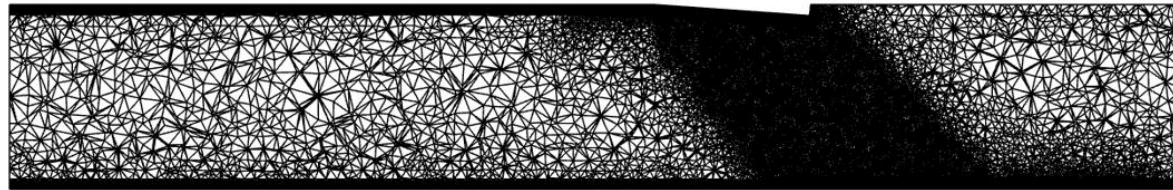
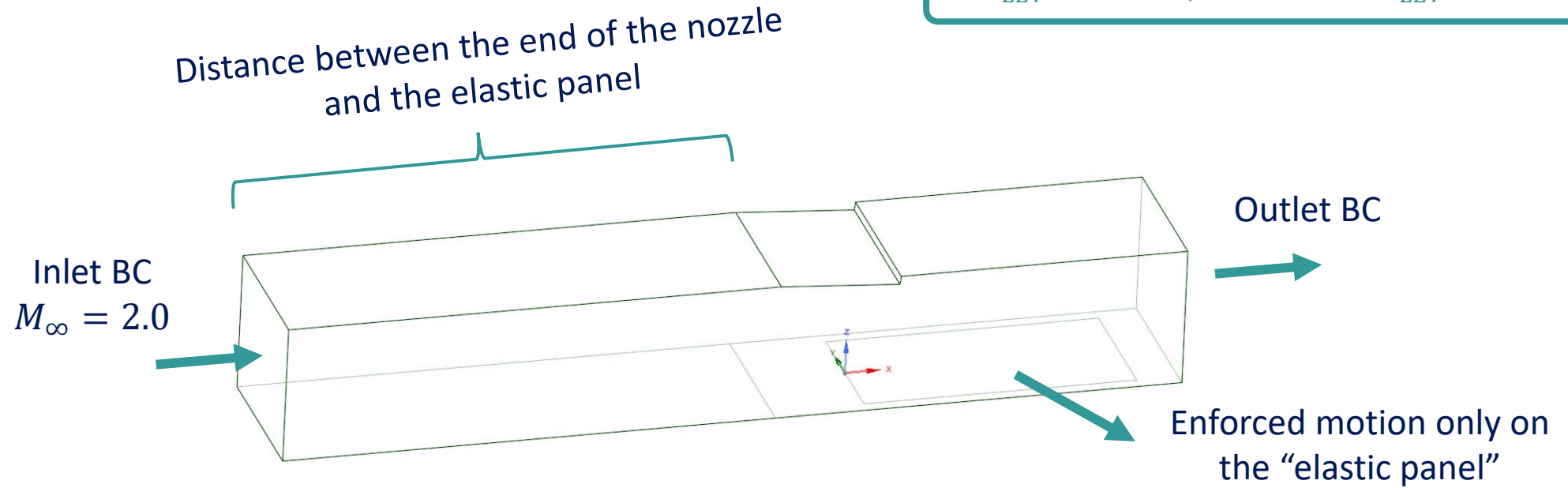
ΔT between panel and frame

Δp between fluid and acoustic cavity

DLTA with a Shock Impingement

$\theta = 4^\circ$ wedge shock configuration: CFD setup

SST $k - \omega$
 $y^+ = 5$ and 1
 $\delta_{LE} \approx 8\text{mm}$ (Wind Tunnel: $\delta_{LE} \approx 8.6\text{mm}$)
 $\delta_{LE}/a \approx 0.03$ (Wind Tunnel: $\delta_{LE}/a \approx 0.033$)



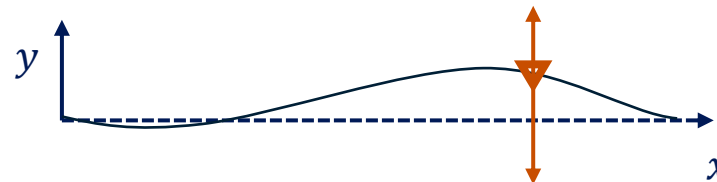
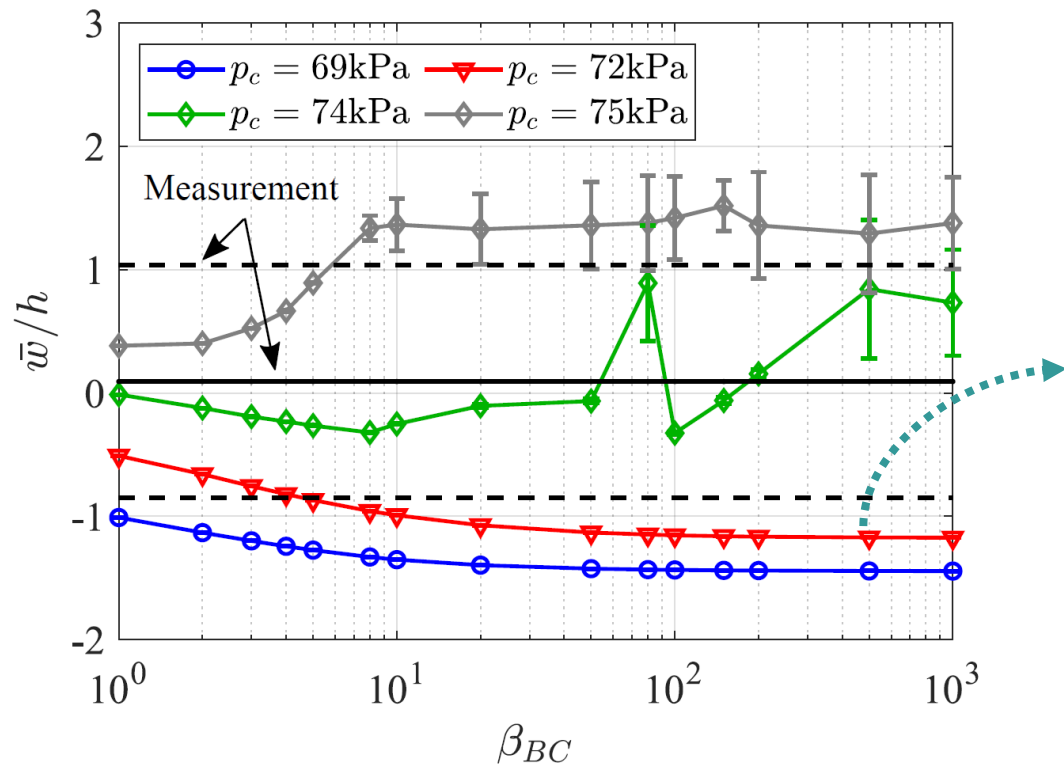
Inflation layer on the lower and upper (pre-wedge/wedge) walls

$\theta = 4^\circ$ wedge shock configuration

$$\Delta T = 15 \text{ K}$$

$$y^+ = 1$$

Mean & SDT deformation (at $\frac{3}{4}$ of the panel length)



Wind Tunnel Setting

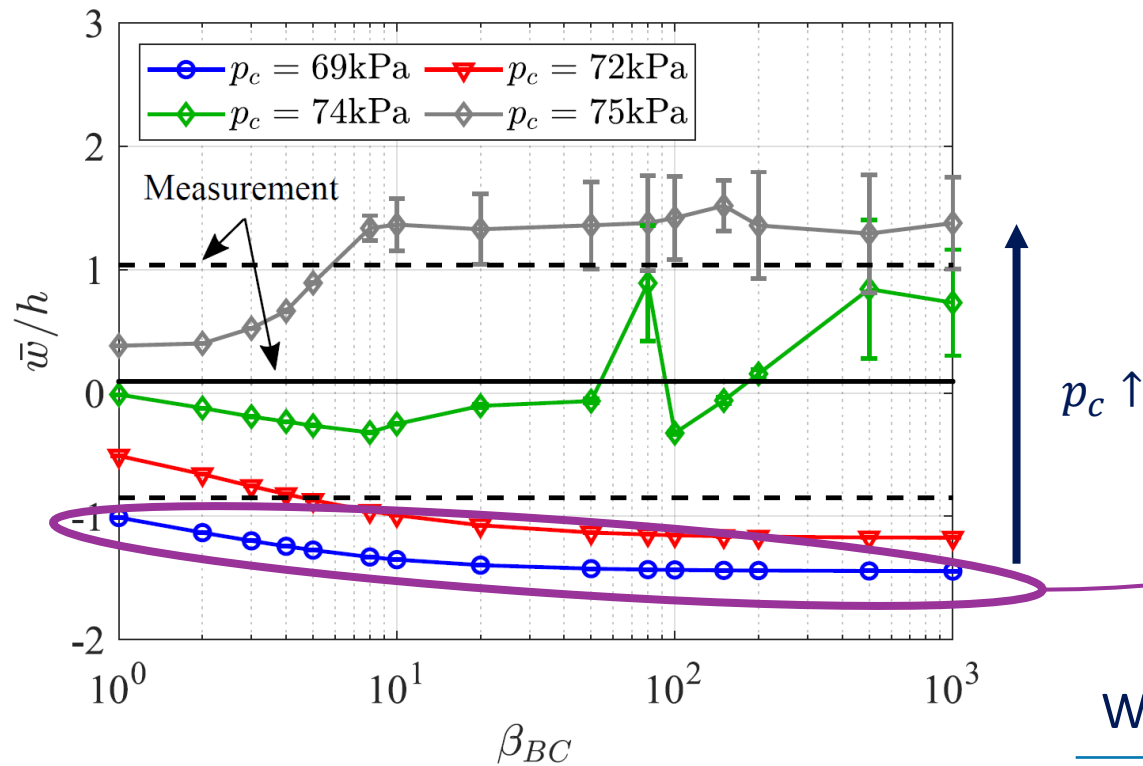
p_c (kPa)	ΔT (K)
68.9	13.3

DLTA with a Viscous Shock Impingement

$\theta = 4^\circ$ wedge shock configuration

$$\Delta T = 15 \text{ K}$$
$$y^+ = 1$$

Mean & SDT deformation
(at $\frac{1}{4}$ of the panel length)



Similar to the wind tunnel
setting

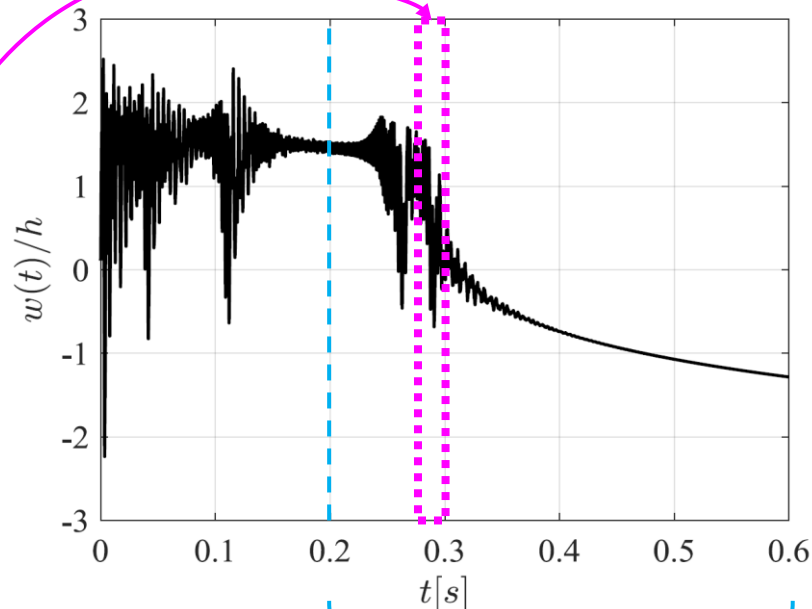
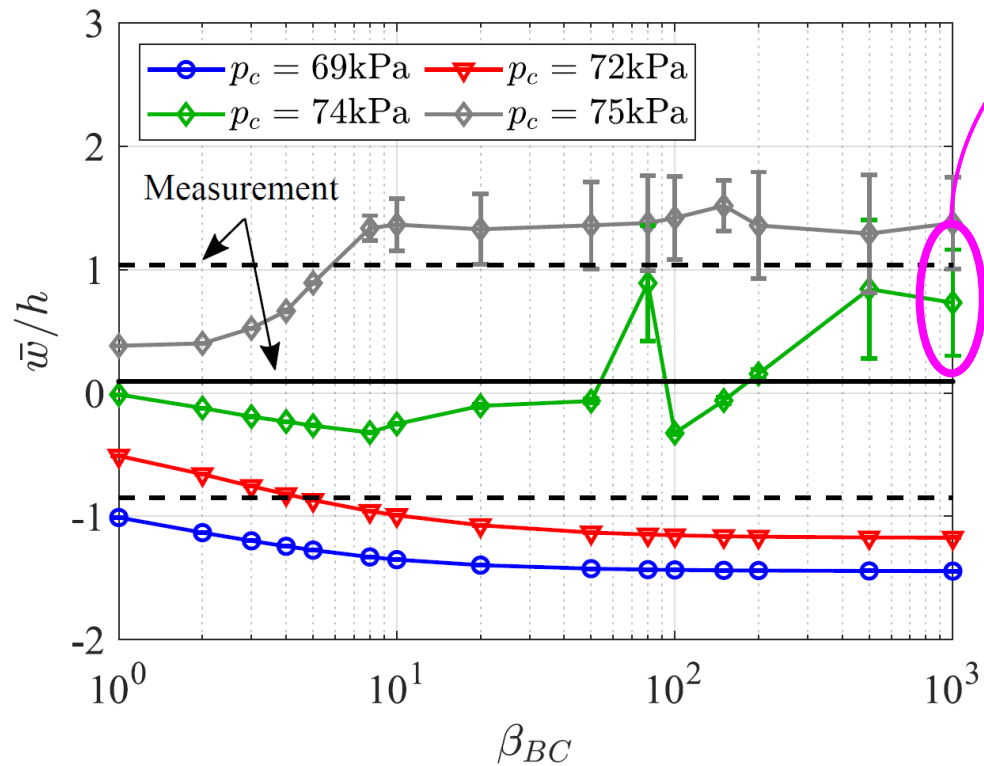
Wind Tunnel Setting

p_c (kPa)	ΔT (K)
68.9	13.3

DLTA with a Viscous Shock Impingement

$\theta = 4^\circ$ wedge shock configuration

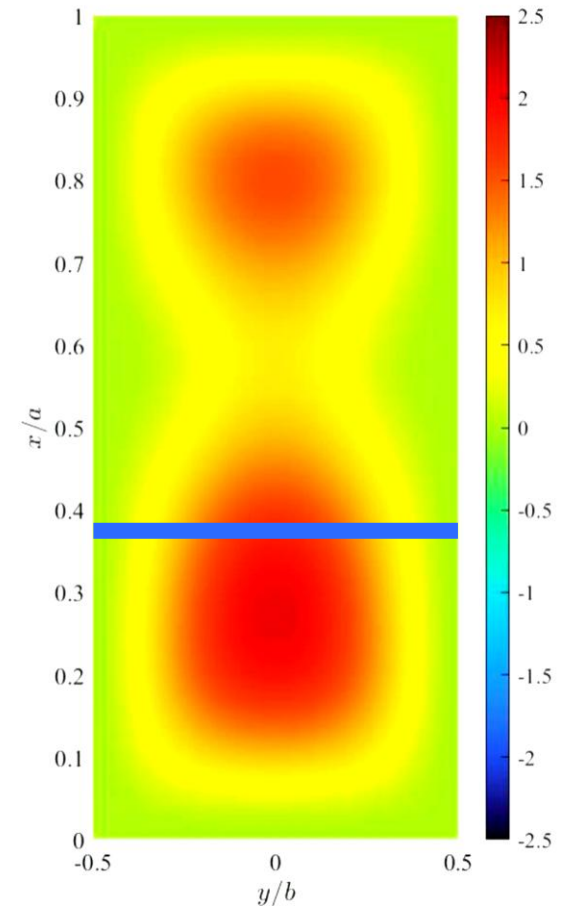
Mean & SDT deformation
(at $\frac{1}{4}$ of the panel length)



Wind Tunnel Setting

p_c (kPa)	ΔT (K)
68.9	13.3

$\Delta T = 15$ K
 $y^+ = 1$

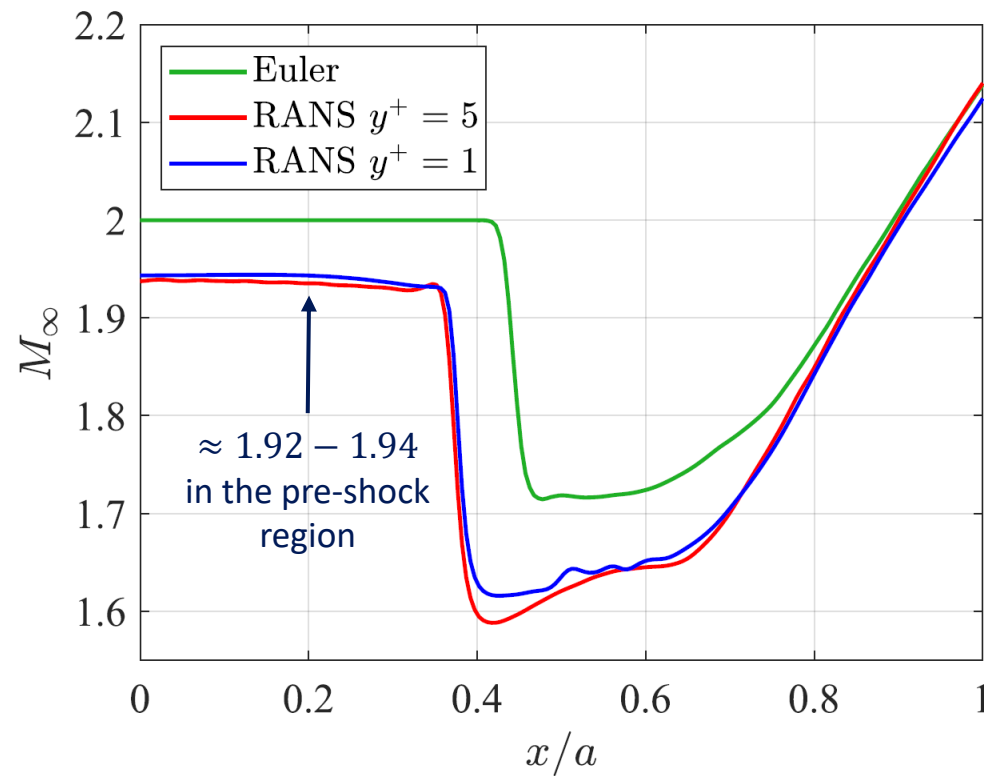


Flow Field

Shock Impingement Aerodynamic Properties

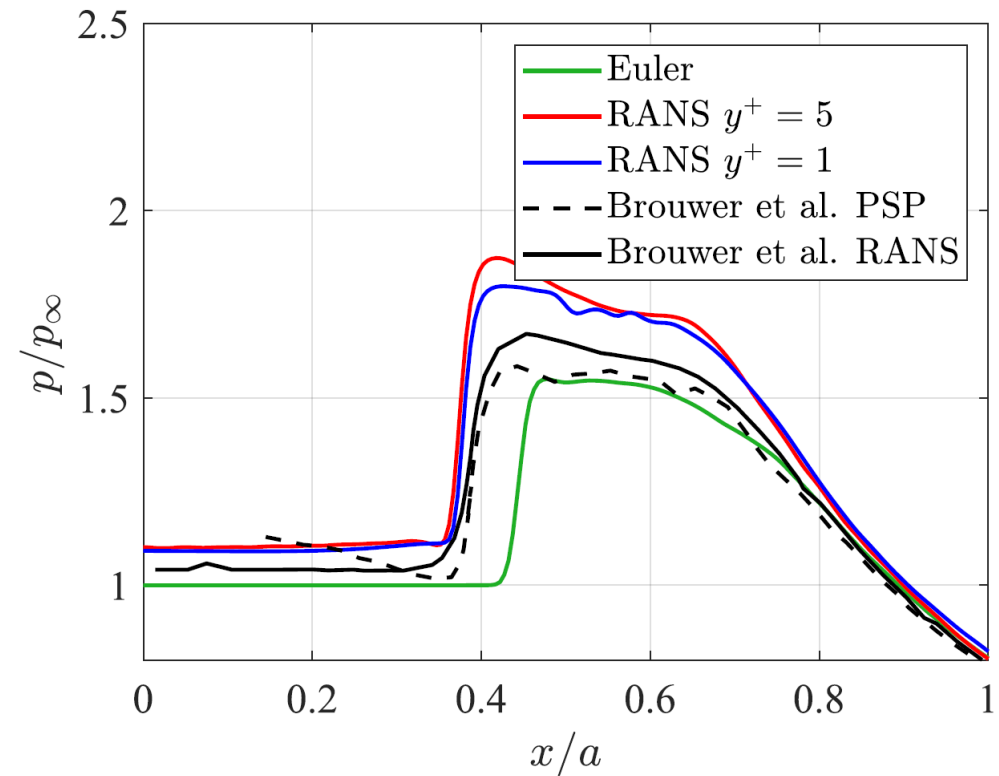
Steady Solution

Local Mach number



Flow more likely to cause structural instability

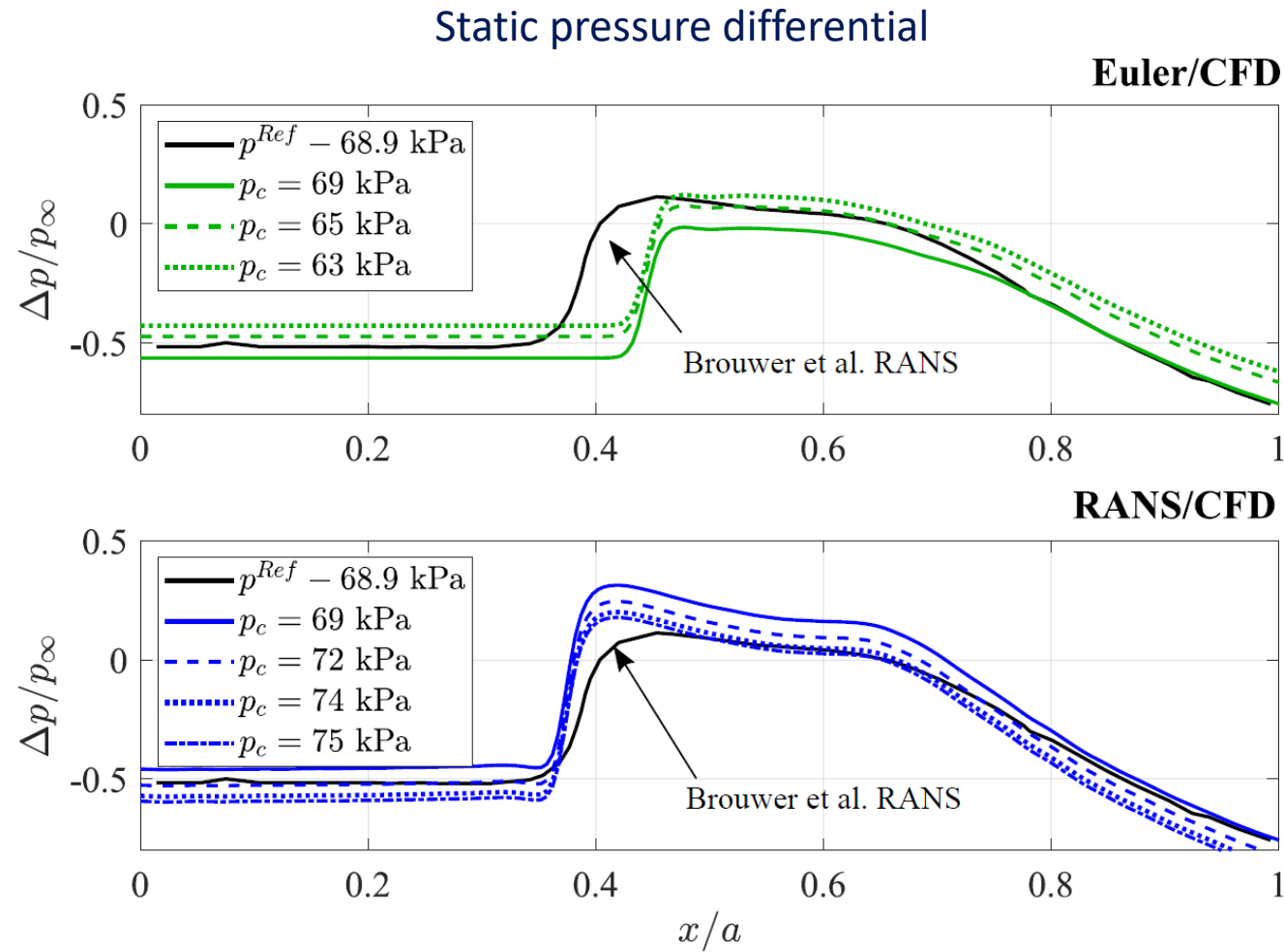
Static pressure



Panel stiffened by the Δp effect

Shock Impingement Aerodynamic Properties

Steady Solution



Panel stiffened by the Δp effect

Conclusion

- A range of aerodynamic models has been considered including Linear Piston Theory and Full Potential Flow for the no-shock case, and Euler and RANS/DLTA for the shock impingement case.
- For the RC-19 configuration **the results are particularly sensitive to the pressure differential, thermal stress (which leads to buckling) and the in-plane as well as out of plane boundary support conditions** for the plate.
- **Results for flutter and LCO of the RC-19 experiment are not particularly sensitive to the aerodynamic model**, with the key exception that when using the DLTA/CFD method, the steady solution for the shock location/magnitude can present indirect implications in the LCO prediction.

The computational models agree with the observations from experiments **on the essentials of the physical phenomena** e.g. buckling, flutter and limit cycle oscillations.

There is broad quantitative agreement between computations and experiments, given the sensitivity of the results to a wide range of parameters.

Correlation of Mathematical Model with AFRL RC-19 Aerothermoelastic Experiment Including an Impinging Shock

AIAA AVIATION Forum 2025

Luisa Piccolo Serafim and Earl Dowell – Duke University

Kirk Brouwer – AFRL, Wright-Patterson AFB

July 22, 2025 | Las Vegas, NV